Applied Linear Algebra

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March 5, 2023

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Outline

Norm

Distance

Angle

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Norm

- The norm of a vector $\mathbf{u}$ in $\mathbb{R}^2$ is given by:

$$||u|| = \sqrt{u_1^2 + u_2^2}$$

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- A unit vector is a vector who norm is 1, e.g: $\frac{u}{||u||}$ is a unit vector

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For any n-vectors $\mathbf{u}$ and $\mathbf{v}$, and any scalar c:

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- Homogeneity: $||c\mathbf{u}|| = |c|||\mathbf{u}||$
- Triangle inequality: $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$
- Nonnegativity: $||\mathbf{u}|| \geq 0$
- Definiteness: $||\mathbf{u}|| = 0$ only if $\mathbf{u} = 0$

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- Euclidean distance:

$$\text{dist}(\mathbf{u}, \mathbf{v}) = ||\mathbf{u} - \mathbf{v}||$$

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Example: Feature distance and nearest neighbors

- If $\mathbf{u}$ and $\mathbf{v}$ are feature vectors for two entities, $||\mathbf{u} - \mathbf{v}||$ is the feature distance

- If $\mathbf{v_1}, \ldots, \mathbf{v_m}$ is a list of vectors, $\mathbf{v_j}$ is the nearest neighbor of $\mathbf{u}$ if for all $i = 1, \cdots, m$:

$$||\mathbf{u} - \mathbf{v_j}|| \leq ||\mathbf{u} - \mathbf{v_i}||$$

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Cauchy- Scwharz inequality

- $|\mathbf{u} \cdot \mathbf{v}| \leq ||\mathbf{u}||||\mathbf{v}||$
- Written also as:

$$|u_1 v_1 + \ldots u_n v_n| \leq \sqrt{(u_1^2 + \ldots + u_n^2)(v_1^2 + \ldots + v_n^2)}$$

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- the cosine of the angle θ between two nonzero vectors $\mathbf{u}$, $\mathbf{v}$ defined as
$$\cos(\theta) = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} \right)$$
- the anlge θ is also denoted by $\angle(\mathbf{u}, \mathbf{v})$
- $\angle(\mathbf{u}, \mathbf{v})$ is between $[0, \pi]$

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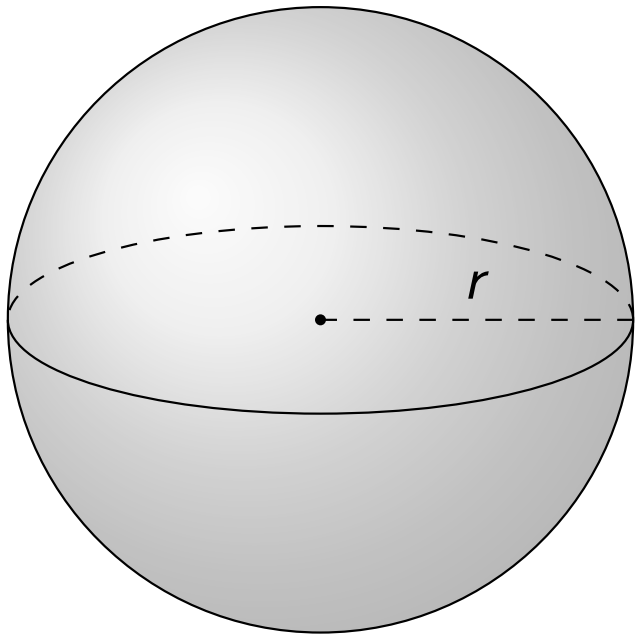
Classification of angles

$\theta = \angle(\mathbf{u}, \mathbf{v})$

- $\theta = \pi/2 = 90°$, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, written $\mathbf{u} \perp \mathbf{v}$, $(\mathbf{u} \cdot \mathbf{v} = 0)$
- $\theta = 0$, $\mathbf{u}$ and $\mathbf{v}$ are aligned, $(\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}||)$
- $\theta = \pi = 180°$, $\mathbf{u}$ and $\mathbf{v}$ are anti-aligned, $(\mathbf{u} \cdot \mathbf{v} = -||\mathbf{u}||||\mathbf{v}||)$
- $\theta \leq \frac{\pi}{2}$, $\mathbf{u}$ and $\mathbf{v}$ make an acute angle, $(\mathbf{u} \cdot \mathbf{v} \geq 0)$
- $\theta \geq \frac{\pi}{2}$, $\mathbf{u}$ and $\mathbf{v}$ make an obtuse angle, $(\mathbf{u} \cdot \mathbf{v} \leq 0)$

Spherical distance

if $\mathbf{u}, \mathbf{v}$ are on sphere of radius r, distance along the sphere is $r\theta$